

STA 013 — Elementary Statistics

Problem Set 5

*Discrete Random Variables, Expected Value & Variance;
Summation Notation; Measures of Center*

Based on Lecture: June 30 (Day 5) • Professor Max

Name: Student ID:

Section: Date:

Instructions: This problem set covers material from Day 5 lecture **only**: probability distributions of discrete random variables, mean and variance of a random variable, summation notation, and the median/mode/mean–median skew relationship. Show **all** work in the space provided for full credit on written-response items. For multiple-choice questions marked with a **Justify** line, a correct letter with no supporting work will receive partial credit only. Calculators are permitted; unsimplified fractions are acceptable unless otherwise noted.

Part I — Probability Distributions, Mean & Variance

1. A discrete random variable x has the following partially completed probability distribution table: [4 pts]

x	1	2	3	4
$P(x)$	0.10	0.25	?	0.35

- (a) Find the missing value of $P(x = 3)$.
 (b) Explain, using a rule from lecture, *why* your method in part (a) is guaranteed to work for any probability distribution table, no matter how many values of x there are.

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2. Multiple Choice. A fair four-sided die (faces labeled 1–4) is rolled once. Let x be the number rolled. Which of the following is the correct expected value E_x ? [3 pts]

(A) $E_x = 2.0$

(B) $E_x = 2.5$

(C) $E_x = 4$

(D) $E_x = 1/4$

Justify your answer using the formula $E_x = x_1P(x_1) + x_2P(x_2) + \cdots + x_nP(x_n)$:

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3. Worked Problem. A game show contestant spins a wheel divided into 3 unequal regions. The random variable x = amount won (in \$), with the distribution below: [8 pts]

x (dollars won)	0	10	50
$P(x)$	0.50	0.40	0.10

- (a) Verify this is a valid probability distribution.
- (b) Compute μ_x , the mean amount won per spin.
- (c) Compute σ_x^2 using the *shortcut formula* from lecture: $\sigma_x^2 = \sum x_i^2P(x_i) - \mu_x^2$.
- (d) In one to two sentences, interpret μ_x in the context of this game (i.e., what does it mean about the “average” outcome of playing many times?).

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4. Worked Problem — Fair Coin, 3 Flips. A fair coin is flipped three times. Let x = the number of heads obtained. [10 pts]

- (a) List all $2^3 = 8$ equally likely outcomes of the sample space, and record the value of x for each.
- (b) Construct the probability distribution table for x (values 0, 1, 2, 3).
- (c) Compute μ_x .
- (d) Compute σ_x^2 using the shortcut formula.

5. Multiple Choice. Which of the following is the correct relationship between the two variance formulas presented in lecture? [3 pts]

- (A) The conceptual formula $\sum (x_i - \mu)^2 P(x_i)$ and the shortcut formula $\sum x_i^2 P(x_i) - \mu_x^2$ give *different* numerical answers, but the shortcut is close enough for exam purposes.
- (B) The two formulas are algebraically identical — they will always produce the exact same numerical value for the same data.
- (C) The shortcut formula only works when $\mu_x = 0$.
- (D) The conceptual formula is used for populations, and the shortcut formula is used for samples.

Part II — Summation Notation

6. Let $x_1 = 3$, $x_2 = -2$, $x_3 = 5$, $x_4 = -1$. Compute each of the following. Show every intermediate step — do not skip to the final number. [12 pts]

(a) $\sum_{i=1}^4 x_i$

$$(b) \sum_{i=1}^4 x_i^2$$

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$$(c) \left(\sum_{i=1}^4 x_i \right) - 4$$

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$$(d) \sum_{i=1}^4 (x_i - 4)$$

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7. Concept check. Using your answers to Question 6, parts (c) and (d): are the two results the same number or different numbers? In two to three sentences, explain *why* the placement of “−4” inside versus outside the summation symbol produces the result you found. [5 pts]

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8. Multiple Choice. Given $x_1 = 2$, $x_2 = 4$, $x_3 = 6$, what is $\sum_{i=1}^3 (5 - x_i)^2$? [4 pts]

- (A) 6
- (B) 9
- (C) 11
- (D) 14

Show your work:

Part III — Median, Mode, and Skewness

9. Worked Problem. Consider the data set: 12, 5, 9, 9, 21, 14 [9 pts]

- (a) Sort the data in ascending order.
- (b) Using $\text{Position} = 0.5(n + 1)$, find the position of the median. Show your arithmetic.
- (c) Using the midpoint rule (if needed), find the median.
- (d) Identify the mode of this data set.

10. Worked Problem. Consider the data set: 8, 3, 8, 11, 3, 8, 3 [6 pts]

- (a) Find the median.
- (b) Find the mode(s). Is this data set unimodal or bimodal? Justify using the definition given in lecture.

11. Multiple Choice. A histogram of a data set shows a long tail of large values extending to the right, with most of the data clustered on the left near a single peak. Based *only* on this shape description (no calculation needed), which relationship below must hold? [4 pts]

- (A) Mean = Median
- (B) Mean > Median
- (C) Mean < Median
- (D) Cannot be determined without the exact data values

Justify your answer in one to two sentences, referencing why the shape produces this relationship:

12. Multiple Choice. A data set is known to be strongly **left-skewed**. A classmate calculates the mean to be 42 but cannot remember whether they calculated the median correctly as 38 or as 47. Using the skew rule from lecture, which value is the more plausible median? [4 pts]

- (A) 38, because Mean < Median must hold for left-skewed data
- (B) 47, because Mean < Median must hold for left-skewed data
- (C) 38, because Mean > Median must hold for left-skewed data
- (D) 47, because Mean > Median must hold for left-skewed data

Justify:

13. Short answer. A dataset is perfectly symmetric and unimodal. A student argues: "Since the mean and median are equal here, they must always be equal for any data set." Explain, in three to four sentences, what is wrong with this reasoning, and describe the specific condition (from lecture) under which mean and median are *not* equal. [5 pts]

Part IV — Mixed Review & Synthesis

14. Synthesis Problem. A small bag contains 4 chips labeled 1, 2, 3, and 3 (i.e., two chips share the label “3”). One chip is drawn at random, and x is the number on the chip. [12 pts]

(a) Construct the probability distribution table for x . (*Hint: there are only 3 distinct values of x , but the probabilities are not all equal.*)

(b) Compute μ_x .

(c) Compute σ_x^2 using the shortcut formula.

(d) Without drawing a picture, state whether you would expect this distribution to be symmetric, right-skewed, or left-skewed, and briefly justify using the shape of the probabilities you found in part (a).

15. Multiple Choice. Which statement correctly distinguishes *sampling variation* from a fixed population value, as discussed at the start of Day 5’s Chapter 2 material? [3 pts]

- (A) Population values are random; sample values are fixed.
- (B) Both population values and sample values are random.
- (C) Population values are fixed; sample values are random.
- (D) Neither population nor sample values can be described as fixed or random — the distinction does not apply.

End of Problem Set 5 — Continue to Answer Key on the following page

Answer Key & Full Solutions

Problem Set 5 — Day 5 Material

1. (a) Since all probabilities in a distribution must sum to 1:

$$0.10 + 0.25 + P(x = 3) + 0.35 = 1 \implies P(x = 3) = 1 - 0.70 = 0.20$$

(b) This works for *any* number of values of x because a probability distribution table lists every possible outcome of the random variable, and the events are mutually exclusive and exhaustive — so by the fundamental rule from lecture, the relative frequencies (probabilities) of all outcomes must sum to exactly 1, no matter how many rows the table has.

2. **Answer: (B)** $E_x = 2.5$

$$E_x = 1\left(\frac{1}{4}\right) + 2\left(\frac{1}{4}\right) + 3\left(\frac{1}{4}\right) + 4\left(\frac{1}{4}\right) = \frac{1+2+3+4}{4} = \frac{10}{4} = 2.5$$

3. (a) $0.50 + 0.40 + 0.10 = 1.00$ ✓ Valid distribution.

(b) $\mu_x = 0(0.50) + 10(0.40) + 50(0.10) = 0 + 4 + 5 = 9$

(c) $\sum x_i^2 P(x_i) = 0^2(0.50) + 10^2(0.40) + 50^2(0.10) = 0 + 40 + 250 = 290$

$$\sigma_x^2 = 290 - (9)^2 = 290 - 81 = 209$$

(d) On average, across many repeated spins, a contestant would win \$9 per spin. This does *not* mean any single spin pays out \$9 — it is the long-run average.

4. (a) Outcomes: HHH(3), HHT(2), HTH(2), THH(2), HTT(1), THT(1), TTH(1), TTT(0)

(b) Distribution table:

x	0	1	2	3
$P(x)$	1/8	3/8	3/8	1/8

(c) $\mu_x = 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right) = 0 + \frac{3}{8} + \frac{6}{8} + \frac{3}{8} = \frac{12}{8} = 1.5$

(d) $\sum x_i^2 P(x_i) = 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 4\left(\frac{3}{8}\right) + 9\left(\frac{1}{8}\right) = 0 + \frac{3}{8} + \frac{12}{8} + \frac{9}{8} = \frac{24}{8} = 3$

$$\sigma_x^2 = 3 - (1.5)^2 = 3 - 2.25 = 0.75$$

5. **Answer: (B)** The shortcut formula is derived algebraically from the conceptual formula by expanding $(x_i - \mu)^2$ — they are not approximations of each other, they are the exact same quantity written two different ways. This mirrors the coin-flip example from lecture, where both formulas gave $\sigma_x^2 = 1/2$ for two flips.

6. (a) $\sum x_i = 3 + (-2) + 5 + (-1) = 5$

(b) $\sum x_i^2 = 3^2 + (-2)^2 + 5^2 + (-1)^2 = 9 + 4 + 25 + 1 = 39$

(c) $(\sum x_i) - 4 = 5 - 4 = 1$

(d) $\sum (x_i - 4) = (3 - 4) + (-2 - 4) + (5 - 4) + (-1 - 4) = (-1) + (-6) + (1) + (-5) = -11$

7. **Different numbers:** part (c) = 1, part (d) = -11. In part (c), the constant 4 is subtracted only *once*, after all four values have already been added together. In part (d), the constant 4 is subtracted from *each individual* x_i before summing — so it is effectively subtracted four times (once per term), not once. This is exactly the inside-vs-outside distinction flagged in lecture: $\sum (x_i - c) \neq (\sum x_i) - c$ in general.

8. Answer: (C) 11

$$(5 - 2)^2 + (5 - 4)^2 + (5 - 6)^2 = 3^2 + 1^2 + (-1)^2 = 9 + 1 + 1 = 11$$

9. (a) Sorted: 5, 9, 9, 12, 14, 21

(b) Position = $0.5(n + 1) = 0.5(6 + 1) = 0.5(7) = 3.5$

(c) Position 3.5 falls between the 3rd and 4th sorted values (9 and 12). Midpoint rule:

$$\text{Median} = \frac{9 + 12}{2} = \frac{21}{2} = 10.5$$

(d) Mode = 9 (it appears twice; every other value appears once).

10. (a) Sorted: 3, 3, 3, 8, 8, 8, 11. Position = $0.5(7 + 1) = 4$. The median is the 4th value: 8.

(b) Both 3 and 8 occur three times each — more than any other value, and tied with each other. By the lecture definition (a distribution with two modes is bimodal), this data set is **bimodal**, with modes 3 and 8.

11. **Answer: (B) Mean > Median.** A cluster on the left with a long tail to the right is, by definition, **right-skewed**. Per the skew rule from lecture, the tail of large extreme values on the right pulls the *mean* upward (since the mean is sensitive to magnitude), while the median — based on rank position, not size — stays near the cluster. Hence Mean > Median for right-skewed data.

12. **Answer: (B) 47, because Mean < Median must hold for left-skewed data.** The rule from lecture states Skewed Left $\implies$ Mean < Median. Since the mean is given as 42, the median must be *larger* than 42 for this rule to hold. Of the two candidate values, only $47 > 42$, so 47 is the plausible median.

13. The student's reasoning commits the error of generalizing from a single special case. Mean equals median *only* under symmetric, unimodal distributions — this is one specific shape, not a universal law. From lecture, whenever a distribution is skewed (right or left), the long tail of extreme values pulls the mean away from the median in the direction of the tail, while the median remains anchored near the bulk of the data by rank rather than magnitude. So the condition under which mean $\neq$ median is precisely when the distribution is skewed rather than symmetric.

14. (a) $P(x = 1) = \frac{1}{4}$, $P(x = 2) = \frac{1}{4}$, $P(x = 3) = \frac{2}{4} = \frac{1}{2}$ (two of the four chips are labeled 3).

x	1	2	3
$P(x)$	$1/4$	$1/4$	$1/2$

(b) $\mu_x = 1(\frac{1}{4}) + 2(\frac{1}{4}) + 3(\frac{1}{2}) = \frac{1}{4} + \frac{2}{4} + \frac{6}{4} = \frac{9}{4} = 2.25$

(c) $\sum x_i^2 P(x_i) = 1(\frac{1}{4}) + 4(\frac{1}{4}) + 9(\frac{1}{2}) = \frac{1}{4} + \frac{4}{4} + \frac{18}{4} = \frac{23}{4} = 5.75$

$$\sigma_x^2 = 5.75 - (2.25)^2 = 5.75 - 5.0625 = 0.6875$$

(d) The probability mass is concentrated more heavily at the larger value ($x = 3$, with $P = 0.5$) than at the smaller values, so we'd expect the distribution to lean **left-skewed** in shape — more weight sits on the right side ($x = 3$) with comparatively less mass trailing off to the left ($x = 1, 2$).

15. **Answer: (C).** Population values are fixed — they don't change depending on who observes them. Sample values are random, because which sample you happen to draw (and therefore which values you get) varies by chance from one sample to the next.

Study tip: For any skew-rule question (like Q11–Q12), always re-derive the inequality from scratch using the rule table (Symmetric: Mean = Median; Skewed Right: Mean > Median; Skewed Left: Mean < Median) rather than trying to memorize which letter was correct on a past problem — the numbers and skew direction will change on the exam.